

Maternal Plasma Cell-Free RNA as a Predictor of Early and Late-Onset Preeclampsia Throughout Pregnancy

Corresponding Author: Dr Tamara Garrido-Gómez

This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have satisfactorily addressed my concerns

Reviewer #2

(Remarks to the Author)

I appreciate the authors' efforts in carefully addressing each one of my questions. I am fine with most of the explanations, revisions and updates provided. However, I am quite alarmed with the lack of stratification of the groups based on aspirin prophylaxis. Low dose aspirin is more effective in mitigating early onset preeclampsia (EOPE). Yet, 30 of the 42 EOPE cases and 23 of the 43 late onset cases had aspirin. In particular, the EOPE cohort of cases may represent a (? particularly prone) group that would develop PE despite aspirin. Yet, some "controls" might have developed PE if not for the aspirin. Instead, only 10 of the 131 controls had aspirin. Thus, I am not certain what the cRNA profiles are reflecting due to the heterogenous nature of the cases and controls. I believe to address this properly, an even larger sample group separating those EOPE/LOPE/controls with and without aspirin prophylaxis to do the cRNA mining separately. At present, I do not know how the study findings could be applied clinically. Do we apply the test to those had or have not had aspirin? In the worst-case scenario, a test subject on aspirin would much more likely to be identified as PE because in the current study cohort, the selected control cases had low aspirin usage.

Reviewer #3

(Remarks to the Author)

The authors have made improvements to their manuscript. In particular the inclusion of the flow chart has helped clarify the design clearly. I also applaud the authors for including the external validation set, although I believe some further information is needed regarding this data set given it is quite small in size (particularly for the 2nd trimester samples).

I have some further specific queries/suggestions for the authors.

The abstract needs to be updated to include the new validation data generated from the external data set; preferably noting the reduced sensitivity/specificity and small numbers that formed this validation.

Were there differences in gene signatures between those with aspirin administration versus those without?

The authors have skimmed over the data explaining the external validation and should include these results properly under the results section, in the same manner they have explained their original data (eg provide the sensitivity, specificity etc in the results, rather than the reader having to go look for it in the figures/tables).

Can the authors please provide further insight into why the numbers are so variable for this Validation 2 cohort (as shown in supp table 6)? Perhaps providing more detail around this in the section commencing at line 498 would be beneficial.

Although the authors say at line 262 Further validation using an independent external dataset confirmed its robustness and translational potential of our predictive signature – in fact, the sensitivities (and specificity) have dropped quite off quite a bit in the validation 2 cohort; possibly because the numbers in those studies are very small. For example, based on this Validation 2 data, I would question whether the conclusion at line 303 is accurate. Similar comment is relevant to the paragraph at line 341. I think it would be helpful if the authors revisit these statements. Also, some further information around the differences/similarities of the samples in validation 2 vs this primary study would be helpful. Was it also a case control design etc.

Given the data shown by Validation 2, I think it is important to include some discussion (in the discussion) around the limitations of using the case control design over the case cohort design. This is important because while the authors suggest in their response that a case control design is critical for discovery, in fact, this design may result in the generation of models that are overly optimistic that may be more likely to fail once population studies are undertaken.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have satisfactorily addressed my concerns

Reviewer #2

(Remarks to the Author)

Thank you for having seriously considered the relevance of my comments. I have reviewed the additional data analysis and do not have any other questions to raised.

Reviewer #3

(Remarks to the Author)

Thank you to the authors for the edits that have improved the manuscript.

Can the authors also include the data for the T1 LOPE in the abstract? This is important to clearly show the challenge of predicting LOPE early in pregnancy.

Perhaps in line 59, after sharing the AUC of 0.90 for T2, the authors could note that T1 performance for predicting LOPE was less favourable with an AUC of 0.68. Then include the validation of this with AUC of 0.63 at line 61, when the T2 data is given, but T1 left off.

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RESPONSE TO REFEREES

Maternal Plasma Cell-Free RNA as a Predictor of Early and Late-Onset Preeclampsia Throughout Pregnancy.

Reviewers' Comments:

We thank the reviewers for their valuable comments and suggestions, which have helped us improve our manuscript.

Editor:

The reviewers confidentially recommend including an outline of your response regarding novelty of the study in the article's introduction. Please highlight the key advantages of your study over previous ones.

A. We thank the reviewers for this helpful suggestion, which enabled us to better highlight the novelty of our study. We have revised the Introduction accordingly to include an outline emphasizing novelty and key advantages of our work over previous ones (lines 101-106).

Reviewer #1:

The authors have satisfactorily addressed my concerns.

A. We thank the reviewer for their positive feedback and are pleased to know that our revised manuscript has satisfactorily addressed all prior concerns.

Reviewer #2:

I appreciate the authors' efforts in carefully addressing each one of my questions. I am fine with most of the explanations, revisions and updates provided. However, I am quite alarmed with the lack of stratification of the groups based on aspirin prophylaxis. Low dose aspirin is more effective in mitigating early onset preeclampsia (EOPE). Yet, 30 of the 42 EOPE cases and 23 of the 43 late onset cases had aspirin. In particular, the EOPE cohort of cases may represent a (? particularly prone) group that would develop PE despite aspirin. Yet, some "controls" might have developed PE if not for the aspirin. Instead, only 10 of the 131 controls had aspirin. Thus, I am not certain what the cRNA profiles are reflecting due to the heterogenous nature of the cases and controls. I believe to address this properly, an even larger sample group separating those EOPE/LOPE/controls with and without aspirin prophylaxis to do the cRNA mining separately. At present, I do not know how the study findings could

be applied clinically. Do we apply the test to those had or have not had aspirin? In the worst-case scenario, a test subject on aspirin would much more likely to be identified as PE because in the current study cohort, the selected control cases had low aspirin usage.

A. We thank the reviewer for raising this important point regarding aspirin prophylaxis, a recommended intervention for the prevention of preeclampsia, and we appreciate the opportunity to further clarify how this variable was considered in the design and interpretation of our study.

This study was conducted under a non-interventional observational framework. As such, it captures the real-world heterogeneity of clinical decision-making of preventive strategies, including variability in aspirin recommendation timing of initiation, adherence, and discontinuation — all of which critically impact aspirin's effectiveness. Thus, any assumptions regarding treatment efficacy or resistance would be speculative and fall outside the scope of this study. Given this context, stratifying participants solely based on recommendation status would risk introducing substantial bias and may not accurately reflect the true biological impact of aspirin exposure. To clarify this point in the manuscript, we have revised the text to indicate that aspirin was *prescribed* (not administered) (line 151). Additionally, cfRNA samples from the first trimester (T1) were collected before the initiation of any aspirin prophylaxis, ensuring that early model development was unaffected by potential treatment-related transcriptional changes.

While aspirin prophylaxis was not a primary focus of our investigation, we recognize the importance of evaluating whether it could confound cfRNA-based predictions. To address this, we performed a series of dedicated and complementary analyses to assess its potential impact on both transcriptomic profiles and model outputs. These included principal component analysis, differential expression testing, and classifier score comparisons, as detailed below.

(i) Principal component analysis (PCA): To evaluate whether aspirin recommendation was associated with large-scale shifts in cfRNA expression profiles, we conducted PCA using normalized counts of all detected cfRNA transcripts (Figure 1; for reviewer only). Separate analyses were performed for EOPE, LOPE, and control groups at both the first (T1) and second (T2) trimester timepoints. Across all groups and timepoints, samples did not cluster by aspirin recommendation status. Individuals with and without aspirin recommendation appeared intermixed in PCA space, indicating no discernible transcriptomic separation attributable to aspirin exposure at the time of sampling.

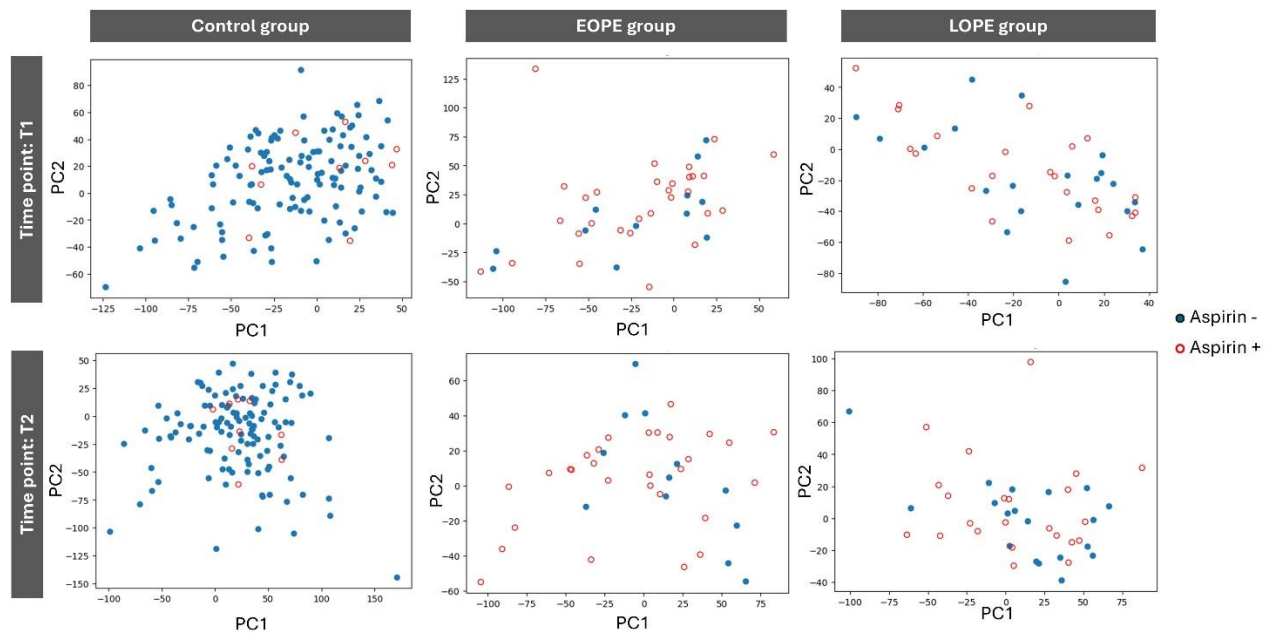


Figure 1 (for reviewer only). Principal component analysis of cfRNA profiles by aspirin prophylaxis recommendation. PCA plots are shown for each group—controls, early-onset preeclampsia (EOPE), and late-onset preeclampsia (LOPE)—at two timepoints: first trimester (T1, top row) and second trimester (T2, bottom row). Each point represents an individual sample. Blue-filled circles indicate participants for whom low-dose aspirin was not recommended. Red-bordered open circles indicate participants who were recommended aspirin prophylaxis.

(ii) Differential abundance analysis: We next performed within-group differential expression analyses, comparing individuals with vs. without aspirin recommendation in each obstetric outcome (EOPE, LOPE, and controls) at both T1 and T2 (Figure 2 for reviewer only). These analyses were carried out using limma-Voom from Bioconductor package, applying an adjusted p-value of 0.05. No statistically significant differentially abundant cfRNA were identified in any group or timepoint. This result held true regardless of gestational age or disease status, suggesting that aspirin recommendation status was not associated with transcriptional differences in cfRNA.

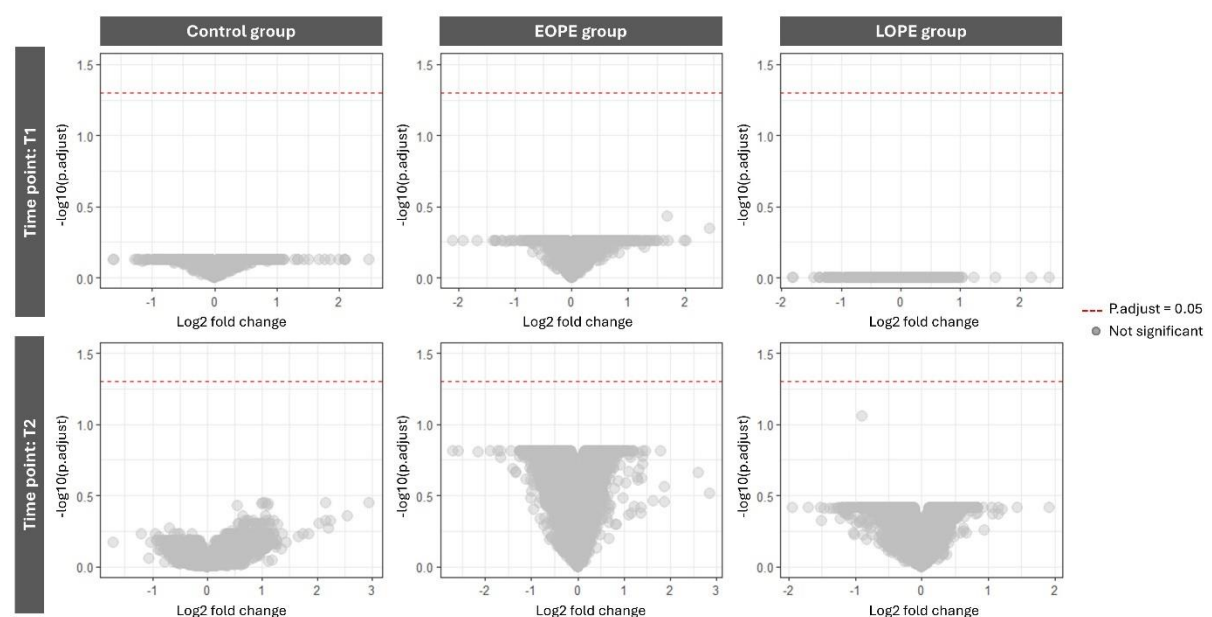


Figure 2 (for reviewer only). Volcano plots showing within-group differential expression analyses based on clinical recommendation for aspirin. Columns represent clinical groups (Control, EOPE, LOPE), and rows indicate gestational windows (Top: T1; Bottom: T2). No significantly differentially expressed genes were detected in any comparison (P -value adjust < 0.05). Grey dots represent non-significant cfrNAs. The red horizontal line marks the significance threshold (P -value adjust $= 0.05$).

(iii) Model output comparison: Classifier score distributions were also comparable across aspirin-recommended and non-recommended individuals (Figure 3 for reviewer only), with no systematic shift in predictions or increased misclassification rates. Statistical comparisons using the Mann-Whitney U test confirmed no significant differences between groups, with p -values reported for each subgroup (see figure legend for details).

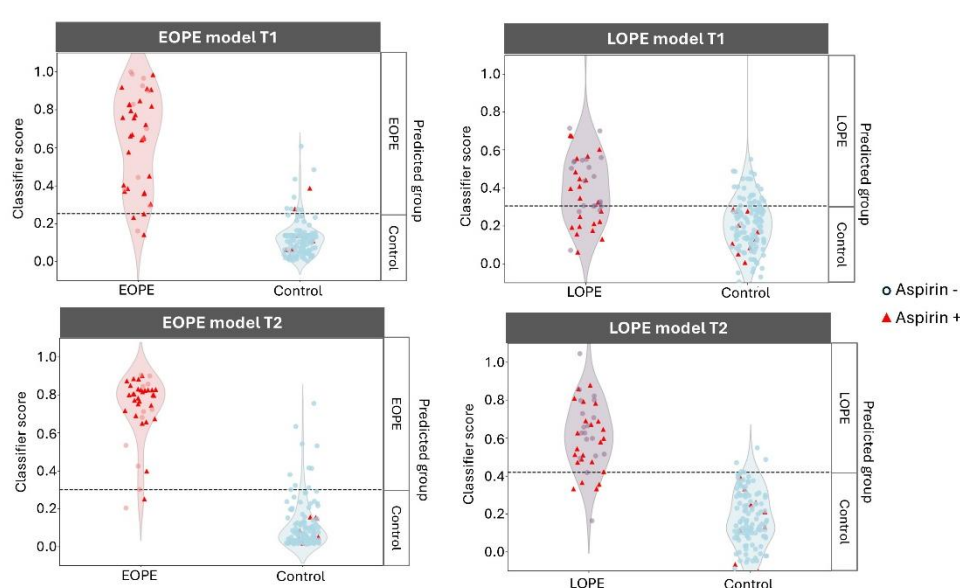


Figure 3 (for reviewer only). Violin plots displaying classifier score distributions stratified by aspirin recommendation status for EOPE and LOPE at both T1 and T2. Each panel shows the distribution of predicted scores for true cases and controls, as assigned by the corresponding predictive model. The Y-axis represents the classifier score, and the X-axis indicates the actual clinical outcome (EOPE/LOPE or control). Individuals recommended aspirin are shown in red-outline triangles, while those not recommended aspirin are shown in circles. Dashed lines indicate the classification threshold used to distinguish predicted cases from controls. Statistical comparison of classifier scores between aspirin-recommended and non-recommended individuals showed no significant differences, with Mann-Whitney U test p -values as follows:

EOPE model T1: EOPE group, $p = 0.30$; control group, $p = 0.36$

LOPE model T1: LOPE group, $p = 0.28$; control group, $p = 0.25$

EOPE model T2: EOPE group, $p = 0.37$; control group, $p = 0.75$

LOPE model T2: LOPE group, $p = 0.23$; control group, $p = 0.87$

Taken together, these findings indicate that aspirin recommendation did not introduce detectable transcriptomic variation in cfrNA profiles nor compromise model performance at either timepoint. These results corroborates that aspirin prophylaxis did not act as a confounding factor in our analyses. Accordingly, the cfrNA signatures observed in EOPE and LOPE reflect the underlying disease biology, independent of aspirin prophylaxis.

Regarding clinical applicability, the cfrNA-based test we propose is intended for integration into routine pregnancy screening, analogous to current first-trimester risk assessment strategies. It is designed to operate independently of prior aspirin exposure and may support clinical decision-making both by identifying

individuals at high risk prior to prophylaxis (at T1) and by flagging persistent or emerging risk later in gestation (at T2), even among those who may have received preventive treatment.

Finally, we emphasize that this study represents an important step toward clinical implementation, but not a finalized product. A large-scale external validation is currently underway as part of the iPregnostic study (ClinicalTrials.gov Identifier: NCT06716242), which is specifically designed to assess model performance across a broader range of clinical settings and real-world conditions. This next phase will further establish the generalizability and clinical utility of cfRNA-based prediction.

Reviewer #4:

The authors have made improvements to their manuscript. In particular the inclusion of the flow chart has helped clarify the design clearly. I also applaud the authors for including the external validation set, although I believe some further information is needed regarding this data set given it is quite small in size (particularly for the 2nd trimester samples).

A. We appreciate your statement about the improvement of our work.

I have some further specific queries/suggestions for the authors.

1) The abstract needs to be updated to include the new validation data generated from the external data set; preferably noting the reduced sensitivity/specificity and small numbers that formed this validation.

A1: We have incorporated the validation data from the external dataset in the abstract (lines 56-66). To accommodate this new information while keeping the abstract concise and within the journal's word limit, we revised the presentation of the model performance metrics, simplifying some details to enhance clarity and readability.

2) Were there differences in gene signatures between those with aspirin administration versus those without?

To address if there were differences in gene signature between aspirin recommendation vs without, we performed a series of dedicated and complementary analyses to assess its potential impact on both transcriptomic profiles and model outputs. These included principal component analysis, differential expression testing, and classifier score comparisons, as detailed below.

(i) Principal component analysis (PCA): To evaluate whether aspirin recommendation was associated with large-scale shifts in cfRNA expression profiles, we conducted PCA using normalized counts of all detected cfRNA transcripts (Figure 1; for reviewer only). Separate analyses were performed for EOPE, LOPE, and control groups at both the first (T1) and second (T2) trimester timepoints. Across all groups and timepoints, samples did not cluster by aspirin recommendation status. Individuals with and without aspirin recommendation appeared intermixed in PCA space, indicating no discernible transcriptomic separation attributable to aspirin exposure at the time of sampling.

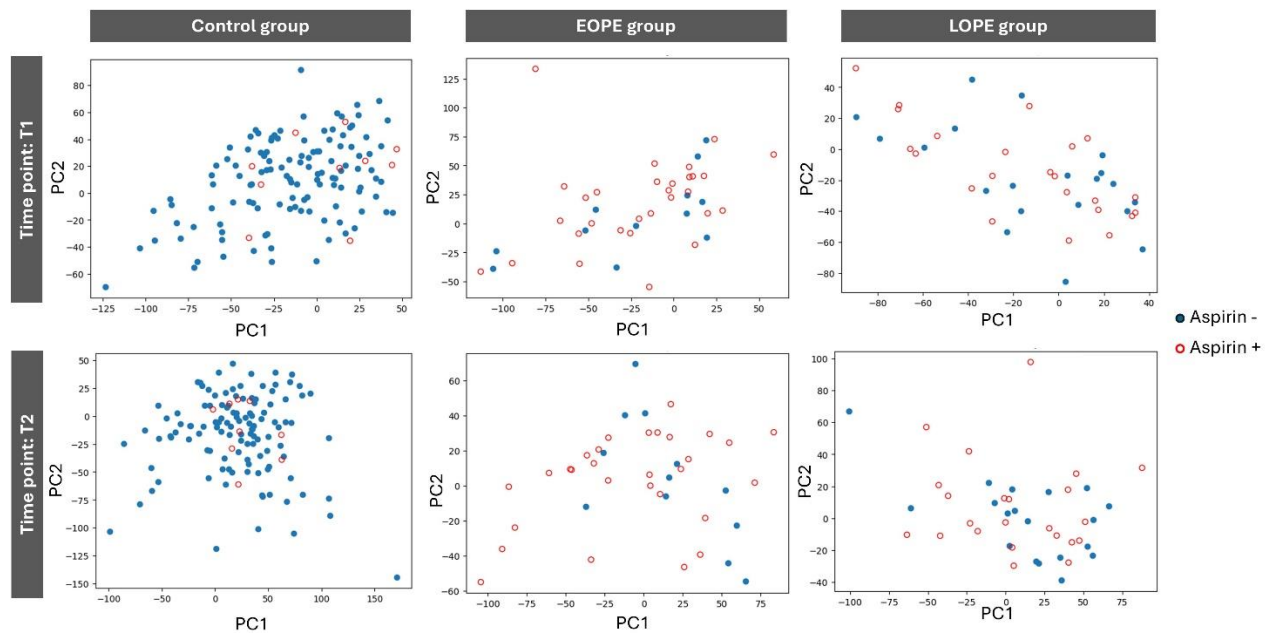


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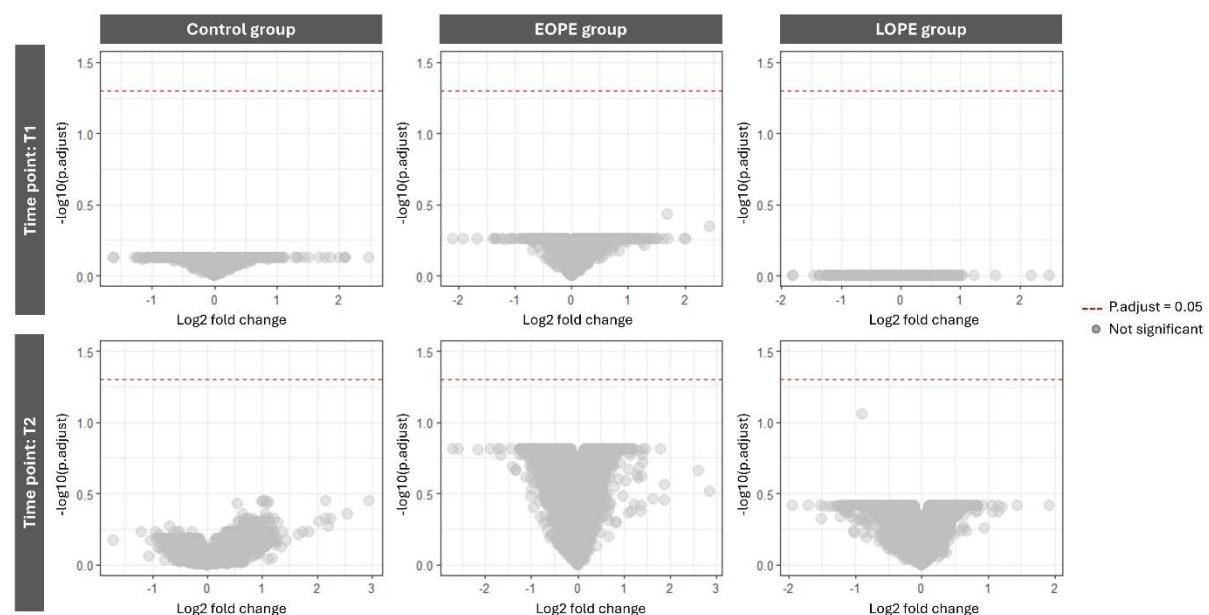


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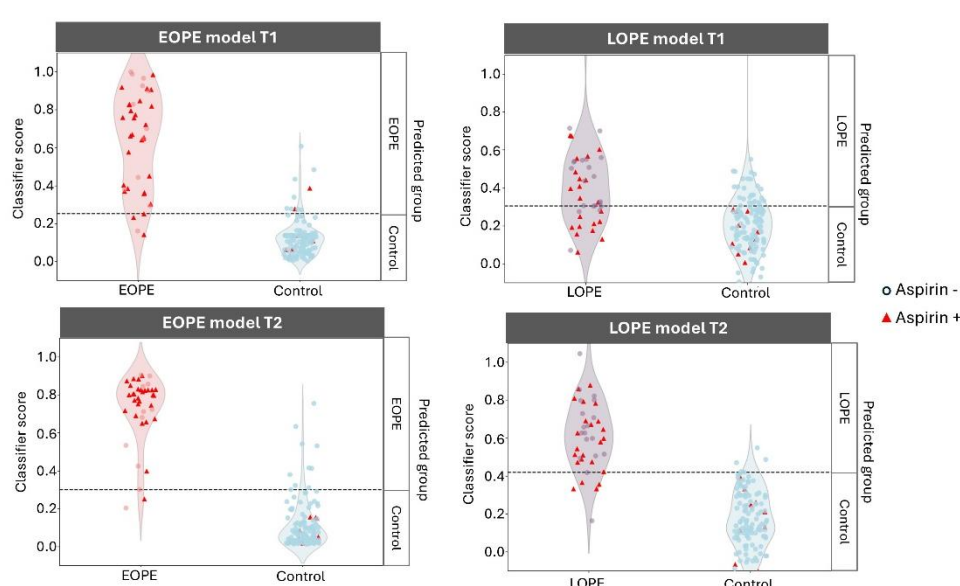


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3) The authors have skimmed over the data explaining the external validation and should include these results properly under the results section, in the same manner they have explained their original data (eg provide the sensitivity, specificity etc in the results, rather than the reader having to go look for it in the figures/tables).

A3: Thank you very much for your valuable comment. We have now revised the Results section to include a clear and concise presentation of the external validation metrics, providing sensitivity, specificity, and AUC values alongside the internal validation results for both EOPE and LOPE models (lines 224-229, 238-239, 258-260; 276-280). Additionally, we have included an explanation for the observed moderate reduction in performance in the external dataset, highlighting factors such as sample timing and metadata limitations that affect direct comparability. We believe these clarifications improve the clarity of the manuscript.

4) Can the authors please provide further insight into why the numbers are so variable for this Validation 2 cohort (as shown in supp table 6)? Perhaps providing more detail around this in the section commencing at line 498 would be beneficial.

A4: We thank the reviewer for pointing out the variability in sample numbers within the Validation 2 cohort (GSE192902, Supplementary Table 6). This variability arises from several factors related to the external origin of the dataset, including differences in study design, metadata availability, and technical protocols.

Specifically, to validate our cfRNA signatures in this independent dataset, we retuned the model architecture to address technical differences in input features—stemming from cfRNA extraction, library construction, sequencing and data processing—which required balanced case–control distributions for each subtype.

Since the original study analysed all preeclampsia cases as a single group without differentiating between EOPE and LOPE, we constructed balanced case–control sets separately for each subtype to align with our study framework. All available preeclampsia cases were included. Controls were balanced using an agnostic down sampling approach based on cfRNA profiles across the signature genes and the gestational sampling window (T1, T2). For each analysis, the largest subset of controls was retained using a reproducible selection criterion within each relevant window. Model performance was assessed using leave-one-out cross-validation and evaluated with standard metrics: accuracy, sensitivity, specificity, AUC, and F1-score. This approach allowed us to evaluate the performance of the gene signature in an external context while accounting for both technical and clinical differences between datasets. Additional methodological detail has been added to the manuscript (lines 521-528).

5) Although the authors say at line 262 Further validation using an independent external dataset confirmed its robustness and translational potential of our predictive signature – in fact, the sensitivities (and specificity) have dropped quite off quite a bit in the validation 2 cohort; possibly because the numbers in those studies are very small. For example, based on this Validation 2 data, I would question whether the conclusion at line 303 is accurate. Similar comment is relevant to the paragraph at line 341. I think it would be helpful if the authors revisit these statements. Also, some further information around the differences/similarities of the samples in validation 2 vs this primary study would be helpful. Was it also a case control design etc.

A5: Thank you for this helpful comment. We have carefully revised the manuscript to address these concerns. In particular, we have removed the statement previously found at line 303 and revised the related sections (now lines 276–280 and 358–360). Specifically, we now describe the Validation 2 results as supporting the relevance of the gene signature across datasets, while acknowledging the modest drop in predictive performance, likely due to technical and sampling variability—including differences in cfRNA extraction, sequencing protocols, and clinical metadata granularity.

6) Given the data shown by Validation 2, I think it is important to include some discussion (in the discussion) around the limitations of using the case control design over the case cohort design. This is important because while the authors suggest in their response that a case control design is critical for discovery, in fact, this design may result in the generation of models that are overly optimistic that may be more likely to fail once population studies are undertaken.

A6: We appreciate the reviewer’s point regarding the limitations of a case–control design. While this approach is commonly used in biomarker discovery studies to optimize statistical power and minimize confounding, we agree that it may limit generalizability. We have now added a statement to the Discussion acknowledging this limitation (lines 360–361). Importantly, to address this concern, a large-scale prospective validation study (iPregnostic; ClinicalTrials.gov Identifier: NCT06716242) is currently ongoing to evaluate model performance in broader, real-world populations.

RESPONSE TO REFEREES

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Reviewer #3:

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- A. We appreciate the reviewer's suggestion. We have now included the T1 LOPE data in the abstract as requested. At this final editorial stage, the journal requires the abstract to remain strictly within a 150-word limit, we have complied with that adding your suggested improvements. Thanks for your invaluable help to improve our paper.
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